

Identity dynamics refute the commuting prime-time recurrence formula

CONJECTURE 00000000139 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

System

Use the two-point space $\{0,1\}$ with masses $1/2$, the singleton $A=\{0\}$, and identity transformations. The transformations preserve probability and commute.

All recurrence times

Every inverse image of A under an identity iterate is A . Every two-time or multiple-recurrence intersection therefore has measure $1/2$, whereas the square of the measure is $1/4$.

Averages and limit

Each nonempty normalized average is $1/2$, regardless of which prime or almost-prime times are selected. Assigning empty averages zero gives values only in the closed set $\{0,1/2\}$; no sequence of these values tends to $1/4$. This does not assume infinitely many twin primes.

Interpretation

The filed statement quantifies over all measure-preserving systems, without an ergodicity or mixing assumption. The identity example addresses precisely that quantifier.

Formalization

Lean constructs the actual two-point counting measure scaled by $1/2$, proves total mass one and identity measure preservation, computes all recurrence intersections, and proves the claimed limit impossible. It also specializes to the arithmetic selection in this conjecture; for commuting recurrence it proves the transformations commute.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000139.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.